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A THESIS PROPOSAL ENTITLED

A NEW EPT-G FAMILY OF DISTRIBUTIONS: PROPERTIES,
APPLICATIONS AND MACHINE LEARNING BENCHMARKING

BY
DONKOH BENJAMIN ARKOH

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THESIS SUPERVISORS

.....
ASSOC. PROF JOSEPH ACQUAH

.....
DR ABDULZEID ANAFO

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DECLARATION

I declare that this project work/thesis is my own work. It is being submitted for the Degree of Master of Philosophy in Mathematics in the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted for any degree or examination in any other university.

.....

(Signature of candidate)

.....day of (year)

ABSTRACT

DEDICATION

To my parents, Very Rev Benjamin Donkoh and Mrs Sarah Donkoh.

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I am deeply grateful to my supervisors, Associate Prof. Joseph Acquah and Dr. Abdulzeid Anafo, for their guidance, constructive feedback, and unwavering support throughout this research.

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LIST OF ABBREVIATION

SQL

Structured Query Language

CHAPTER 1

INTRODUCTION

1.1 Background of the Research

Statistical distribution theory has an important role in applied probability and mathematical statistics, providing the theoretical insights for modelling random phenomena in all kinds of fields (Egamov *et al.*, 2024). Ever since the work of Pearson (1895), who proposed a system of frequency curves to model observed data distributions, statisticians have been working on trying to extend and generalise classical distributions to the complicated forms of skew, heavy tails, bimodality, and non-monotone hazard rate functions that researchers have encountered. The development of new families of distributions has therefore become one of the most vibrant research programmes in modern statistics (Pizzutilo, 2012; Lahcene, 2013).

Classical distributions, including the exponential, Weibull, gamma, and log-normal distributions, have served as the basis of statistical modelling for decades. The Weibull distribution, in particular, has been extensively applied in reliability engineering, survival analysis, and failure-time modelling due to its flexible shape parameter and interpretable hazard function (Weibull, 1951; Taketomi *et al.*, 2022). Nevertheless, datasets from fields like finance, epidemiology, and biomedical research frequently exhibit features including heavy right-skewness, non-monotone hazard rates, and multimodal density patterns that traditional distributions fail to represent effectively (Alzaatreh *et al.*, 2013; Suleiman *et al.*, 2024).

A common theme in this study is the addition of a parameter to adjust the tails, peak, and hazard function of resulting distributions. Lee *et al.* (2013) showed that using the Weibull transformer yields families of distributions that can model bathtub-shaped hazard rates, which is particularly important in reliability and survival analysis. Bourguignon *et al.*, (2014) proposed the Weibull-G family, and Cordeiro *et al.*, (2019) introduced the odd Weibull-G family, both offering greater flexibility compared

to their original forms. The exponentiated Weibull distribution, extensively examined by Mudholkar *et al.*, (1995) and Nassar and Eissa (2003), showed that adding just one exponentiation parameter enables the modelling of unimodal, bathtub, and increasing hazard shapes, further highlighting the value of exponentiated distribution families.

Benchmarking statistical distributions and machine learning (ML) has become an increasingly important area, with probabilistic models used in algorithms that quantify uncertainty, make predictions, and support decision-making (Murlidhar and Sayed, 2024). Distributions play a vital role in Bayesian networks and Naive Bayes classifiers, as well as generative models, wherein they are used to model the conditional dependencies and to perform inferences under missing information (Li, 2024). In this case, benchmarking the performance of distributions in ML frameworks means the systematic evaluation of the performance in terms of standardised datasets, metrics (e.g., accuracy, precision, cross-entropy loss) and protocols such as cross-validation and statistical tests to guarantee reproducibility and fair comparisons (Canbek, 2023).

Albelali and Ahmed, (2025) identified gaps associated with benchmarking of the ML model, for example, train-test split or result variance, that may result in unreliable measurements, so it is advisable to conduct many tests to cover the performance distributions. StatEval and other specialist benchmarks, such as StatEval, evaluate the performance of LLMs by evaluating statistical reasoning in fields, such as probability, inference, and ML, demonstrating how distributions and algorithmic effectiveness interact (Lu *et al.*, 2025). In infrastructure, such as sewer pipes, comparison of benchmarking statistical models (e.g., Gompertz) with ML methods (e.g., Random Forests) through such metrics as survival analysis can range the benefits of distributions to predictive maintenance and condition assessment (Caradot *et al.*, 2018).

Investigating the statistical properties of the Exponentiated Power-Transformed-G family and benchmarking its performance against machine learning can provide valuable insights into its practical usefulness for modelling complex datasets, for example, financial data (Bitcoin weekly trading volumes), epidemiological data, and biomedical data (breast cancer survival times), which show the domains in which the

distribution is accurate and economically fit.

1.2 Research Problem

The emergence of generalised families of distributions (G-families) has become a basic development in statistics, allowing scholars to model complex datasets unlike the traditional such as exponential, Weibull, and gamma models (Naudts and Anthonis, 2012; Ocloo *et al.*, 2025). These families, along with their properties, have attracted the attention of researchers due to their computational efficiency in representing lifetime and reliability data. However, as the heterogeneity of datasets spans across fields, such as survival analysis, finance, and engineering, there is an increasing need for more distributional forms that can provide a larger degree of flexibility in modelling such diverse hazard-rate profiles, the presence of pronounced skewness, and various kurtosis (Ampadu and Anafo, 2019; Nasiru, 2018; Chipepa *et al.*, 2025).

Some available generalised distributions struggle to provide closed-form expressions for important statistical properties, such as moments, entropy measures, and hazard rate functions, that are analytically computable and numerically stable. Examples of such families include the Kumaraswamy-G and beta-G families, which have an incomplete beta function in their moment formulation and also complicate the analytical derivation. (Cordeiro and de Castro, 2011; Eugene *et al.*, 2002).

The Weibull distribution is one of the most commonly used lifetime models in reliability and survival analysis (Murthy, *et al.*, 2004). It has two parameters that are not found sufficient in the data that exhibit heavy right-skewness, non-monotone hazard rates, or bimodal density shapes (Mudholkar *et al.*, 1995). The EPT-Weibull sub-model, which uses two extra shape parameters in the Weibull baseline through the EPT generator, provides a possible solution to this modelling incompleteness, although its statistical characteristics, parameter estimation methods, and goodness-of-fit comparisons with other distributions have not been strictly determined in the literature.

Machine learning research methods have brought new advanced density estimation

techniques that provide an attractive alternative to traditional parametric distributions. However, there are also a few rigorous studies on benchmarks, especially in comparative studies between new statistical families and machine learning methods (Sperti et al., 2021). Current benchmarking practices are often deficient; this includes train-test partitioning that is not reliable and the high level of variance in the results, which distorts the true performance of a particular model (Lones, 2024). As a result, there is a need to develop standardised procedures to unify heterogeneous data sets, powerful strategies of cross-validation, and a set of measures, including accuracy and cross-entropy loss, to precisely determine whether new statistical distributions, such as the EPT-G family, can compete with or supplement machine-learning density estimators.

This study seeks to develop an EPT-G family of distribution, establish its theoretical properties, and demonstrate its practical utility through real- world applications and machine learning benchmarking.

1.3 Research Questions

- i. How can the EPT-G family of distributions be mathematically formulated, and what are its fundamental structural properties?
- ii. What specific mathematical characteristics and statistical properties distinguish the EPT-Weibull distribution when derived as a member of the EPT-G family?
- iii. How effectively does the EPT-Weibull distribution model fit into real-world datasets?
- iv. How does the EPT-Weibull model benchmark against modern machine learning density estimators?

1.4 Research Objectives

The objectives of this research are to:

- i. develop the Exponentiated Power Transform—Generalised (EPT-G) Family,

- ii. introduce the Exponentiated Power Transform- Weibull (EPT-W) as a special case and derive its properties,
- iii. apply the Exponentiated Power Transform- Weibull (EPT-W) to real datasets, and
- iv. benchmark the Exponentiated Power Transform-Weibull (EPT-W) against machine learning density estimators.

1.5 Facilities Used

The following are the facilities to be used are;

- i. Internet facilities
- ii. Personal Laptop
- iii. The University of Mines and Technology Library, Essikado
- iv. R Software

1.6 Thesis Organisation

The thesis is structured into five chapters. Chapter 1 provides an introduction, outlining the problem under study, the main aim of the thesis, the specific research objectives, and the overall organization of the report. Chapter 2 reviews relevant literature from the past decade. Chapter 3 details the development of the new EPT-G family of distributions along with their properties, as well as the formulation of the EPT-Weibull distribution and its characteristics. Chapter 4 discusses the results, focusing on the performance of the EPT-Weibull and maximum likelihood (ML) density estimators. Finally, Chapter 5 presents the study's conclusions and offers recommendations.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

In this chapter, some methods of generating extensions or modifications to statistical distributions are discussed, and...

2.2 Methods of Generating Generalised Distributions

There are various ways to develop new distributions. Some of these methods are the Marshall-Olkin method, methods based on differential equations, the transformation method, the quantile function approach, the beta generation approach, the T-X approach, and the methods of adding parameters.

2.2.1 Marshall-Olkin Method

The Marshall-Olkin technique, named after Albert W. Marshall and Ingram Olkin (1997), is a typical method of constructing new probability distributions by refining an underlying model through parameter induction. This method has found significant attention in statistical literature, particularly in reliability theory and survival analysis, due to its ability to maximise the degree of flexibility without sacrificing the ability to perform analysis (Marshall and Olkin, 1997; Helali, 2026). Brango *et al.*, (2024) highlighted that the Marshall-Olkin method was originally formulated using the exponential and Weibull distributions and showed that the resulting family possesses important theoretical properties, including closure under geometric minima and the ability to be extended to bivariate distributions. Subsequent literature developed the overall mathematical characteristics of the Marshall-Olkin family and proposed a wide variety of models using the Marshall-Olkin to model censored data and to apply it to problems in practice. The literature illustrates that the method is especially efficient in reliability and survival analysis since it is able to generate a range of hazard shapes and enhanced model fit (Ghanbari *et al.*, 2022). Later developments

include the Marshall–Olkin generalised exponential, Marshall–Olkin extended Weibull, beta Marshall–Olkin, and Kumaraswamy Marshall–Olkin families, all of which broaden the scope of the original construction and have been applied to data modelling, stress–strength reliability, and time series analysis (Josse, 2011).

2.2.2 Methods of Differential Equations

The systematic development of the systems of probability distributions was greatly promoted by the initial research of Karl Pearson since 1895. The novelty of Pearson was a methodology to obtain probability density functions as solutions to a particular type of differential equations:

$$\frac{1}{f(x)} \frac{df(x)}{dx} = \frac{a + x}{b_0 + b_1x + b_2x^2} \quad (2.1)$$

(Alzaatreh *et al.*, 2013; Nasiru, 2018)

According to this scheme, various probability distributions can be obtained based on the values of the parameters and the nature of the roots of the quadratic term in the denominator (Alzaatreh *et al.*, 2013). This classification was later expanded by Pearson to include Type I to Type XII and thus offered a flexible framework that could be used to model a large skewness and kurtosis. In this system, there are a number of familiar distributions as special cases. An example is that Type I incorporates the beta distribution as an asymmetric form, whereas the continuous uniform distribution is a limiting case (Stavroyiannis, 2016). The other Pearson systems have useful applications, such as Type IV, which can be used in financial modelling to model non-normal data behaviour, and Type V, the inverse Gamma distribution, which is commonly used in Bayesian and reliability applications (Stavroyiannis, 2016; Monthus, 2023).

Simultaneously with the Pearson approach, the Burr system offers a different method of distribution generation. The Burr family is constructed on cumulative distribution functions, as opposed to probability density functions, which provide an alternative structural view (Chakraborty *et al.*, 2022). Burr Type XII is the most common in this family and has been extensively used in reliability engineering,

hydrology and economics. It is especially adaptable in the heavy-tailed behaviour modelling and, subject to parameter constraints, converts to other known distributions like the Lomax or Pareto Type II (Khalil and Ali, 2023), which brings out the centrality of the use of differential equations and functional methods in the construction and classification of probability distributions.

2.2.3 Method of Transformation

Another method of developing probability distributions is the method of transformation, which has developed in response to the shortcomings of early systems like the Pearson and Burr families. This approach builds up new distributions by using monotonic transformations of familiar random variables, especially the standard normal distribution. One further contribution to this framework was made by Norman Lloyd Johnson (1949), who proposed a systematic transformation-based method of creating flexible families of distributions. Johnson proposed that a non-normal random variable X can be transformed into a standard normal variable Z through a monotonic transformation of the general form:

$$Z = \gamma + \delta f\left(\frac{X - \xi}{\lambda}\right), \quad (2.2)$$

where $f(\cdot)$ is a monotonic transformation function, γ and $\delta > 0$ are shape parameters, ξ is a location parameter, and $\lambda > 0$ is a scale parameter. This transformation provides a mechanism through which a wide variety of distributional shapes can be generated from the normal distribution.

The Johnson system is particularly notable because it classifies distributions into three main sub-families depending on the transformation applied: the SL (lognormal system), the SB (bounded system), and the SU (unbounded system). Each system corresponds to a different functional form of $f(\cdot)$, allowing for different support restrictions and tail behaviours.

A key advantage of the Johnson system is its flexibility in approximating a wide range of skewness and kurtosis structures, making it particularly useful in empirical data modelling. It also includes several well-known distributions as special or limiting

cases, such as the normal, lognormal, and logistic distributions (Johnson, 1949; Van-Dorp and Jones, 2020).

2.2.4 Method of Quantile Function

The idea of the lambda distribution was instrumental in the formulation of the quantile-based method of building probability distributions. This was later generalised into the so-called generalised lambda distribution, which is defined in terms of percentile functions (Nasiru, 2018).

Freimer *et al.* (1988) investigated the connection between the Pearson system and the generalised lambda distribution. They found that the Pearson family lacks the logistic distribution, but that the generalised lambda distribution cannot capture all possible combinations of skewness and kurtosis. To overcome these shortcomings, Karian and Dudewicz (2000) proposed an elongated generalised lambda distribution. This extension takes the characteristics of both the generalised lambda and generalised beta distributions. Further research, which uses quantile-based methods derived from the lambda distribution of Tukey, has been presented by Turner and Pruitt (1978), Morgenthaler and Tukey (2000), and Jones (2002).

2.2.5 Beta- Generated Method

Eugene *et al.*, (2002) proposed a generator of the family of distributions using the beta distribution by generating the beta distribution as the generator. This family of distributions may be said to be a generalization of the distribution of order statistics (Jones, 2004). Since the introduction of the beta-generated family of distributions, numerous beta-generated distributions have been suggested in the literature. Among them are beta-normal, beta-Gumbel, beta-Fréchet, beta-exponential, beta-Weibull, beta-Cauchy, beta-exponentiated Pareto, beta-generalised logistic, and beta-extended Weibull (Nasiru, 2018).

The CDF of the beta-generated family of distributions is defined as follows:

$$G(x) = \frac{1}{B(\alpha, \beta)} \int_0^{F(x)} t^{\alpha-1} (1-t)^{\beta-1} dt \quad (2.3)$$

where $\alpha > 0$ and $\beta > 0$ are shape parameters that control the skewness and kurtosis of the resulting distribution, $B(\alpha, \beta)$ is the beta function, and $F(x)$ is the CDF of any continuous random variable.

The flexibility of the family created by the beta-generated resulted in additional generalisations. The Kumaraswamy-generated family proposed by Jones and Cordeiro and de Castro was to replace the beta distribution with the Kumaraswamy distribution, which provides similar bounded support but with a simpler closed-form CDF (Alzaatreh *et al.*, 2013; Alzaghal *et al.*, 2013). Moreover, Alzaghal *et al.* (2013) generalised this idea to the family of distributions based on the T-X family of distributions, which permits the use of any continuous random variable T as a generator, as opposed to being restricted to distributions defined on the interval (Alzaghal *et al.*, 2013). All these methods can equip any researcher with an effective set of tools to model various phenomena in the fields of engineering, medicine, and reliability analysis (Marias *et al.*, 2013).

2.2.6 Method of Adding Parameters

One of the most popular techniques in recent literature in statistical distribution theory is the approach of introducing additional parameters. An example would be the application of this approach by Stavastava (1993), who proposed the exponentiated Weibull distribution which was obtained by the addition of an extra parameter in the classical Weibull model. The cumulative distribution function (CDF) of the exponentiated family can be expressed as

$$H(x; \alpha, \theta) = G(x; \theta)^\alpha, \quad (2.4)$$

where $\alpha > 0$, $x \in \mathbb{R}$, and α serve as an additional shape parameter. This parameter makes the resulting distribution more flexible and able to model a larger number of data behaviours than the baseline distribution.

Exponentiated distributions are particularly advantageous because they retain the fundamental structure of the original model while improving its adaptability. Examples include extensions of the exponential, gamma, Weibull, Gumbel, and Fréchet

distributions. By utilising this formulation, numerous modified versions of classical distributions have been developed (Anafo, 2022).

Similarly, the concept of adding a single parameter to an existing family of distributions has gained considerable attention in statistical research. This strategy has been widely employed to generalise well-known distributions and improve their modeling capabilities (Ahmad *et al.*, 2019). More broadly, this technique underpins the Marshall–Olkin family of distributions, which is based on systematic parameter augmentation (Tahir and Nadarajah, 2015).

2.3 Summary of Literature Review

CHAPTER 3

METHOD USED

3.1 Overview

This chapter presents the concepts on the methods that were used to achieve the objectives of the study. The topics discussed include the principle behind the maximum likelihood estimation, Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm, goodness of-fit tests, information criteria and total time on test (TTT).

CHAPTER 4

THEORETICAL RESULTS

4.1 Overview

The Exponentiated Power Transform Generalised (EPT-G) family of distributions is introduced in this chapter, and the derivation of the Exponentiated Power Transform Weibull (EPT-W) distribution is presented. The series expansion of the probability density function (PDF) of the proposed EPT-W distribution is obtained for analytical development. Furthermore, some important statistical characteristics of the EPT-W distribution, such as quantile function, moments, mean, variance, moment-generating function, skewness, kurtosis, Rényi entropy, mean deviations, order statistics, and stress-strength reliability, are derived and analysed. Moreover, the parameter estimation method for the proposed EPT-W distribution is presented and discussed.

4.2 EPT-G Family of Distribution

Let X be a random variable with cumulative distribution function (CDF), $G(x; \xi)$, and probability density function (PDF), $g(x; \xi)$, where $x \in \mathbb{R}$, and ξ is a vector of parameters in the random variable X . The CDF of the Exponentiated Power Transform-Generalised (EPT-G) family is defined by:

$$F(x; \alpha, \beta, \xi) = \frac{1 - e^{-\alpha[G(x; \xi)]^\beta}}{1 - e^{-\alpha}}, \quad x \geq 0, \quad \alpha > \frac{1}{e}, \quad \beta > 0 \quad (4.1)$$

where:

α and β are two additional shape parameters,

$G(x; \xi)$ be the CDF of any baseline distribution.

Proposition 4.1: The probability density function (PDF), $f(x; \alpha, \beta, \xi)$, is obtained from differentiating the cumulative distribution function $F(x; \alpha, \beta, \xi)$ with respect to x ;

$$f(x; \alpha, \beta, \xi) = \frac{d}{dx} F(x; \alpha, \beta, \xi) \quad (4.2)$$

Substituting equation (4.1) into equation (4.2),

$$f(x; \alpha, \beta, \xi) = \frac{d}{dx} \left(\frac{1 - e^{-\alpha[G(x; \xi)]^\beta}}{1 - e^{-\alpha}} \right)$$

$$f(x; \alpha, \beta, \xi) = \frac{\alpha\beta g(x; \xi)[G(x; \xi)]^{\beta-1} e^{-\alpha[G(x; \xi)]^\beta}}{1 - e^{-\alpha}}, \quad x \geq 0 \quad (4.3)$$

where:

$g(x; \alpha, \beta, \xi)$ is the PDF of the baseline distribution.

4.2.1 Survival Function

Let X be a random variable with CDF $F(x; \xi)$ and PDF $f(x; \xi)$, where $x \in \mathbb{R}$, and ξ is a vector of parameters in the random variable X , by definition the survival function ($S(x; \xi)$) is defined as

$$S(x; \xi) = 1 - F(x; \alpha, \beta) \quad (4.4)$$

Substituting equation (4.1) into equation (4.4)

$$S(x; \alpha, \beta, \xi) = \frac{e^{-\alpha[G(x; \xi)]^\beta} - e^{-\alpha}}{1 - e^{-\alpha}} \quad (4.5)$$

4.2.2 Hazard Rate Function

The hazard rate function ($h(x)$) is the ratio of the PDF and the survival function given by;

$$h(x) = \frac{f(x; \alpha, \beta, \xi)}{S(x; \alpha, \beta)} \quad (4.6)$$

Substituting equation (4.3) and equation (4.5) into equation (4.6)

$$h(x) = \frac{\alpha\beta g(x; \xi)[G(x; \xi)]^{\beta-1} e^{-\alpha[G(x; \xi)]^\beta}}{e^{-\alpha[G(x; \xi)]^\beta} - e^{-\alpha}} \quad (4.7)$$

4.3 Formulation of the Exponentiated Power Transform-Weibull Distribution

Let X be a random variable with cumulative distribution function (CDF), $G(x; \lambda, k)$, and $g(x; \lambda, k)$ be the PDF of the baseline distribution where $x \in \mathbb{R}$, and $x \geq 0$. Then

the CDF and PDF of the Weibull distribution are defined as;

$$G(x; \rho, k) = 1 - e^{-(\rho x)^k}, \quad \rho = \frac{1}{\lambda} \quad (4.8)$$

$$g(x; \lambda, k) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-\left(\frac{x}{\lambda}\right)^k} \quad (4.9)$$

where $\lambda > 0$ is the scale parameter of the distribution, and $k > 0$ is the shape parameter of the distribution. Substituting equation (4.8) into equation (4.1) gives the CDF of EPT-W distribution:

$$F(x; \alpha, \beta, \rho, k) = \frac{1 - e^{-\alpha(1 - e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}}, \quad \rho = \frac{1}{\lambda} \quad (4.10)$$

In Figure (4.1), the CDF graph increases smoothly from 0 toward 1, showing that the cumulative probability increases as x becomes larger. Different values of α, β, λ and k change the steepness and curvature of the graph, indicating that the EPT-Weibull distribution is highly flexible. Some parameter combinations produce rapid increases in the CDF, suggesting early occurrence of failures or events, while others increase gradually, indicating longer survival times. These observations show that the EPT-Weibull distribution can effectively model a wide range of behaviours.

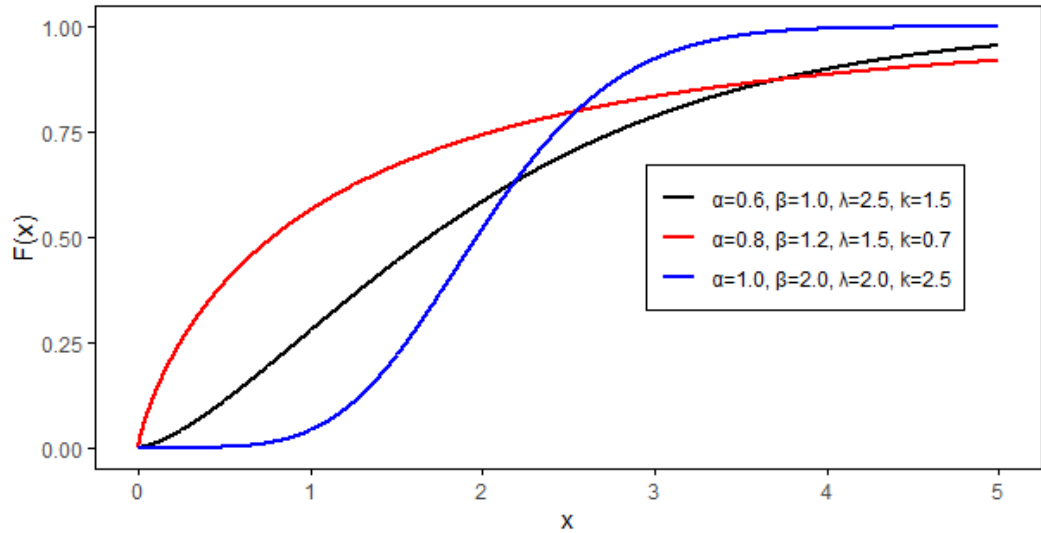


Figure 4.1 CDF Plots of the EPT-W distribution

The PDF of the EPT-W distribution can be obtained by substituting equations (4.8)

and (4.9) into equation (4.3), which gives the EPT-Weibull PDF:

$$f(x) = \frac{\alpha\beta k\rho^k(x)^{k-1}e^{-(\rho x)^k}(1 - e^{-(\rho x)^k})^{\beta-1}e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}}, \quad \rho = \frac{1}{\lambda} \quad (4.11)$$

Figure (4.2) illustrates the sensitivity of the shape of the density function of the distribution to changes in the parameters. For smaller values of α and β , the curve is flatter and more spread out, both indicating low peak density and heavy tails. The larger β and k , the sharper and narrower the peak, and the greater the right skewness; the smaller k , the wider the spread, and the lower the peak. The higher the value of α and β , the more peaked the peak and the more symmetrical its shape is, with a bell shape. Together, these variations illustrate the versatility of the EPT-Weibull family in modelling a variety of data types in applied applications, both heavy-tailed and sharply peaked. The survival function, also the reliability function of the EPT-Weibull

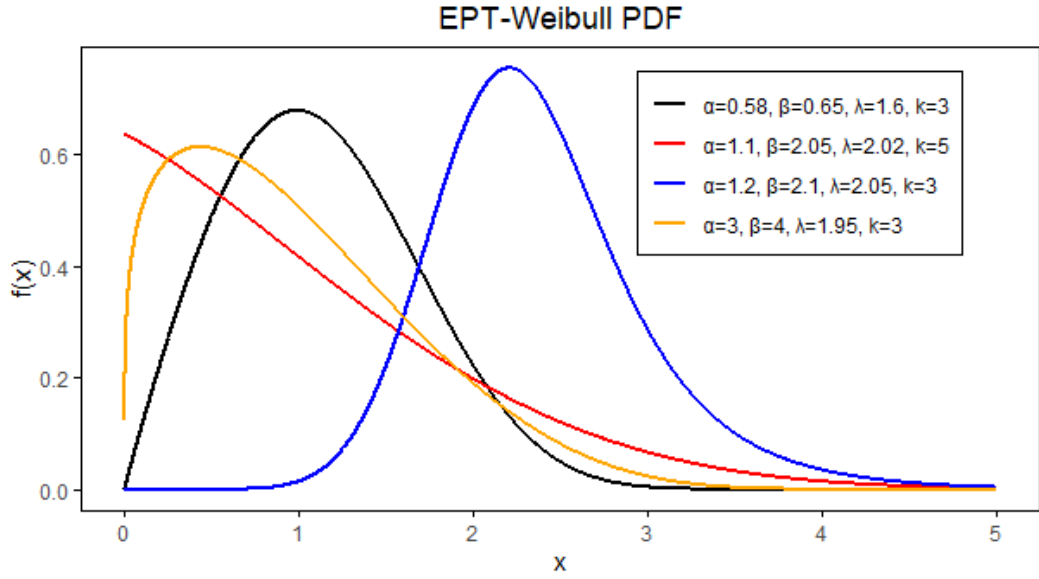


Figure 4.2 PDF Plots of the EPT-W distribution

distribution, is defined as:

$$S(x) = \frac{e^{-\alpha(1-e^{-(\rho x)^k})^\beta} - e^{-\alpha}}{1 - e^{-\alpha}}, \quad \rho = \frac{1}{\lambda} \quad (4.12)$$

Figure (4.3) clearly demonstrates how parameter variations influence the rate of decay and persistence of survival probabilities. Relatively small values of β and k decline

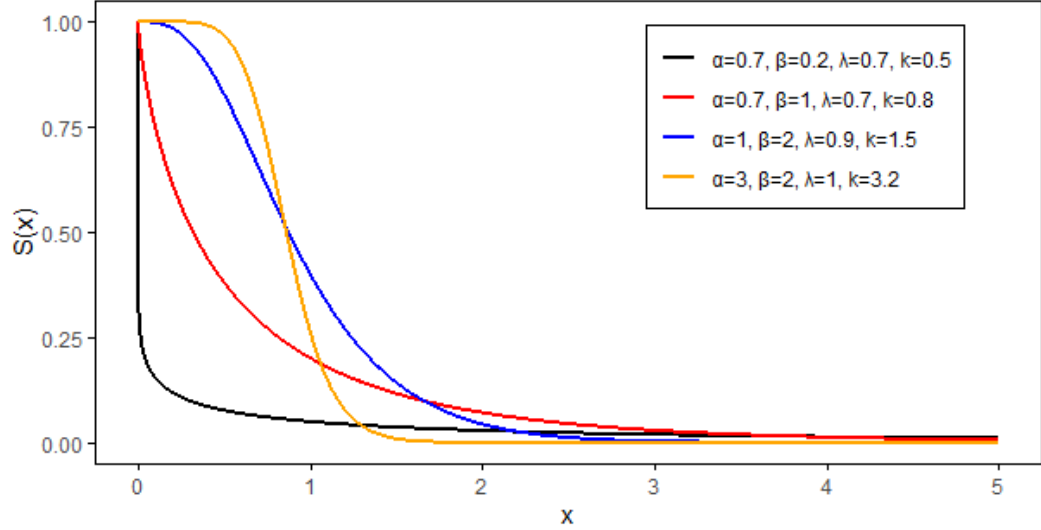


Figure 4.3 Survival function Plots of the EPT-W distribution

more steeply, indicating a faster failure rate and light tail behaviour. With slightly larger β and k , falls more gradually, reflecting a slower decline in survival probability. A higher α, β and k showing a gradual drop, suggesting decreased hazard intensity and increased longevity.

The Hazard function of the EPT-W Distribution is a ratio of the density function to the survival function, which is shown in equation (4.13).

$$h(x) = \frac{\alpha\beta k\rho^k(x)^{k-1}e^{-(\rho x)^k}(1 - e^{-(\rho x)^k})^{\beta-1}e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{e^{-\alpha(1-e^{-(\rho x)^k})^\beta} - e^{-\alpha}}, \quad \rho = \frac{1}{\lambda} \quad (4.13)$$

The hazard function of the EPT-W distribution shown in Figure (4.4) exhibits considerable flexibility under different parameter combinations and shows increasing, approximately constant, and unimodal hazard rate behaviours, indicating that the distribution can effectively model various lifetime characteristics. Some parameter values produce an initially increasing hazard that later stabilises, while others generate a peak followed by a decline or a continuously increasing failure rate. This demonstrates the capability of the EPT-W distribution to model different ageing and reliability patterns commonly encountered in survival analysis and reliability studies.

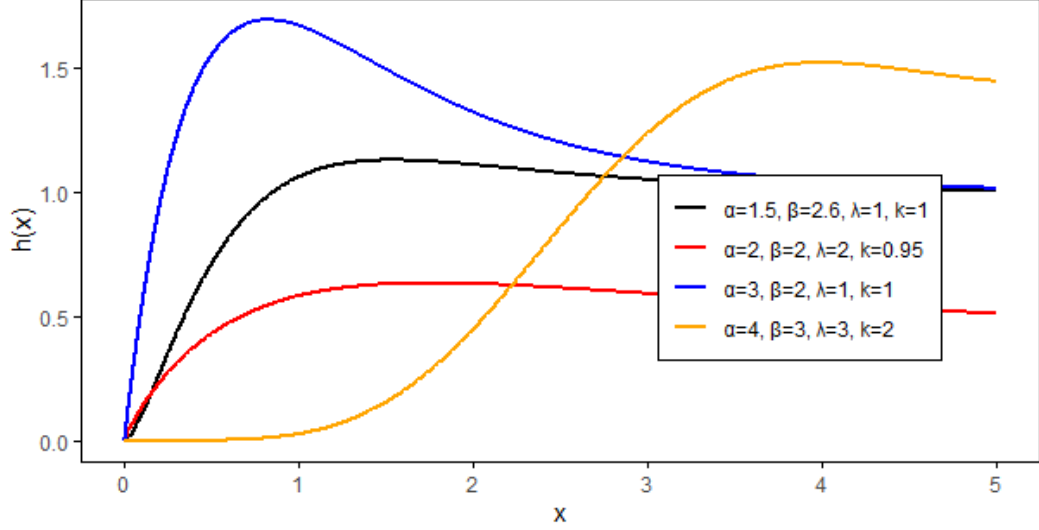


Figure 4.4 Hazard function Plots of the EPT-W distribution

4.3.1 Series Expansion and Mixture Representation of the EPT-W distribution

The density function of the EPT-W distribution is written in a different form to achieve certain statistical characteristics of the distribution. The probability density function of the EPT-W distribution is stated in a mixing form using the concepts of Taylor series expansion and the binomial theorem.

Lemma 4.1 The density function of the EPT-W distribution can be expressed as a series representation.

$$f(x) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} w_{ij} x^{k-1} e^{-(j+1)(\rho x)^k}, \quad \rho = \frac{1}{\lambda} \quad (4.14)$$

where

$$w_{ij} = \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j}$$

Proof: Given the density function,

$$f(x) = \frac{\alpha \beta k \rho^k x^{k-1} e^{-(\rho x)^k} (1 - e^{(\rho x)^k})^{\beta-1} e^{-\alpha(1 - e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}}, \quad \rho = \frac{1}{\lambda}$$

The Taylor series expansion,

$$e^{-m} = \sum_{i=0}^{\infty} \frac{(-1)^i}{i!} m^i$$

is applied on the exponential term, $e^{-\alpha(1-e^{-(\rho x)^k})^\beta}$ where $m = \alpha(1 - e^{-(\rho x)^k})^\beta$ thus

$$e^{-\alpha(1-e^{-(\rho x)^k})^\beta} = \sum_{i=0}^{\infty} \frac{(-1)^i (\alpha)^i}{i!} (1 - e^{-(\rho x)^k})^{\beta i} \quad (4.15)$$

Equation (4.15) is substituted into the density function

$$f(x) = \sum_{i=0}^{\infty} \frac{k\beta\rho^k (-1)^i (\alpha)^{i+1}}{(1 - e^{-\alpha}) i!} (e^{-(\rho x)^k}) (1 - e^{-(\rho x)^k})^{\beta(i+1)-1}, \quad \rho = \frac{1}{\lambda}$$

But the fact that $0 < e^{-(\rho x)^k} < 1$, for $x > 0$ can be further expressed in a series expansion using the binomial expansion, which is,

$$(1 - e^{-(\rho x)^k})^{\beta(i+1)-1} = \sum_{j=0}^{\infty} (-1)^j \binom{\beta(i+1)-1}{j} (e^{-j(\rho x)^k})$$

now, the density function is then expressed as,

$$f(x) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{\beta\rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1)-1}{j} x^{k-1} e^{-(j+1)(\rho x)^k} \quad \rho = \frac{1}{\lambda}$$

which can be written as

$$f(x) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} w_{ij} x^{k-1} e^{-(j+1)(\rho x)^k}, \quad \rho = \frac{1}{\lambda}$$

where

$$w_{ij} = \frac{\beta\rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1)-1}{j}$$

This completes the proof.

4.4 Statistical Properties of the EPT-W distribution

In this section, the structural properties such as the quantile function, skewness and kurtosis, the r^{th} moments, the moment-generating function, the mean, the variance, the entropy, the order statistic, and the strength reliability of the EPT-W distribution are derived and discussed.

4.4.1 Quantile Function

The quantile function is useful for simulating random variates from statistical distributions. The distribution of a random variable can be described using its quantile function. The quantile function is also useful in computing the kurtosis, skewness, and quartiles of a distribution in which the median is one of them.

Proposition 4.2: Given a random variable x with CDF $F(x)$, the quantile function for the EPT-W distribution is given by

$$Q(u) = -\frac{1}{\rho} \left\{ \ln \left[1 - \left(-\frac{1}{\alpha} \ln [1 - u(1 - e^{-\alpha})] \right)^{1/\beta} \right] \right\}^{1/k}, \quad 0 < u < 1. \quad (4.16)$$

Proof: Let u be a random variable which follows the standard uniform distribution, then $0 < u < 1$

$$u = F(x) = \frac{1 - e^{-\alpha(1 - e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}}, \quad 0 < u < 1.$$

Making x the subject, we multiply both sides by $(1 - e^{-\alpha})$ gives

$$u(1 - e^{-\alpha}) = 1 - e^{-\alpha(1 - e^{-(\rho x)^k})^\beta}.$$

Rearranging and taking the natural logarithm on both sides,

$$-\alpha(1 - e^{-(\rho x)^k})^\beta = \ln [1 - u(1 - e^{-\alpha})].$$

Multiplying through by $-\frac{1}{\alpha}$, and raising both sides to the power $\frac{1}{\beta}$, we obtain

$$e^{-(\rho x)^k} = 1 - \left(-\frac{1}{\alpha} \ln [1 - u(1 - e^{-\alpha})] \right)^{1/\beta}$$

Taking natural logarithm again, and making x the subject,

$$(\rho x)^k = -\ln \left[1 - \left(-\frac{1}{\alpha} \ln [1 - u(1 - e^{-\alpha})] \right)^{1/\beta} \right]$$

Therefore,

$$x = \frac{1}{\rho} \left\{ -\ln \left[1 - \left(-\frac{1}{\alpha} \ln [1 - u(1 - e^{-\alpha})] \right)^{1/\beta} \right] \right\}^{1/k}.$$

The quantile function of the distribution is

$$Q(u) = \frac{1}{\rho} \left\{ -\ln \left[1 - \left(-\frac{1}{\alpha} \ln [1 - u(1 - e^{-\alpha})] \right)^{1/\beta} \right] \right\}^{1/k}, \quad 0 < u < 1 \quad \rho = \frac{1}{\lambda}$$

Hence, this completes the proof.

Table (4.1) presents selected quantile values of the EPT-Weibull distribution for varying probabilities u under different parameter settings. The quantiles increase monotonically with increasing u , confirming the validity of the quantile function. Furthermore, the parameter combinations significantly influence the spread and growth rate of the quantiles. In particular, parameter set A exhibits the largest upper quantiles, indicating greater dispersion compared to parameter sets B and C , which display more gradual increases.

Table 4.1 Some values of the quantile of the EPT-W distribution for selected parameter values

$Q(u)$	A	B	C
	$\alpha = 0.5, \beta = 1, \lambda = 1, k = 2$	$\alpha = 1, \beta = 1, \lambda = 1, k = 1$	$\alpha = 2, \beta = 2, \lambda = 1, k = 2$
$Q(0.1)$	0.289297	0.06752788	0.4889531
$Q(0.2)$	0.4231287	0.14521108	0.6069063
$Q(0.3)$	0.5378742	0.23606664	0.7001333
$Q(0.4)$	0.6475381	0.34458678	0.7857715
$Q(0.5)$	0.7592781	0.47785113	0.8714694
$Q(0.6)$	0.8794687	0.64791159	0.9634639
$Q(0.7)$	1.0167633	0.87770671	1.0699951
$Q(0.8)$	1.1877115	1.21944343	1.2070652
$Q(0.9)$	1.4400632	1.84159018	1.4223368

4.4.2 Skewness and Kurtosis

The quantile function is used to find the measure of skewness and kurtosis through the use of Galton's skewness, $\gamma(u)$, defined in Galton (1883), and Bowley's kurtosis, $\kappa(u)$,

also defined in Moors (1988). The measures of Skewness $\gamma(u)$ and Kurtosis $\kappa(u)$ are shown in equations (4.17) and (4.18) respectively.

$$\gamma(\Lambda) = \frac{Q(\frac{3}{4}; \Lambda) + Q(\frac{1}{4}; \Lambda) - 2Q(\frac{2}{4}; \Lambda)}{Q(\frac{3}{4}; \Lambda) - Q(\frac{1}{4}; \Lambda)} \quad (4.17)$$

$$\kappa(\Lambda) = \frac{Q(\frac{7}{8}; \Lambda) - Q(\frac{5}{8}; \Lambda) + Q(\frac{3}{8}; \Lambda) - Q(\frac{1}{8}; \Lambda)}{Q(\frac{6}{8}; \Lambda) - Q(\frac{2}{8}; \Lambda)} \quad (4.18)$$

where Λ is the values of α , β , λ and k

Figures (4.5) and (4.6) are 3D surfaces of Galton's skewness and Moore's kurtosis for the EPT-W distribution for $\alpha = 1$ and $\lambda = 2$. As values of β and k decrease, the skewness ($\gamma(u)$) and kurtosis ($\kappa(u)$) of the distribution increase, suggesting that the distribution is highly skewed and has heavy tails. Also, increasing the values of β and k causes the surfaces to slope downward, making the distribution increasingly symmetric and light-tailed. Both measures converge to near-constant values for k and $\beta \approx 4$, which is a region of Weibull-like symmetry. At low parameter values, the distribution exhibits pronounced asymmetry and heavy tails, while higher values lead to stabilisation and near-symmetric behaviour. This dual control mechanism highlights the flexibility of the EPT-W sub-model in modelling datasets ranging from highly skewed financial returns to more regular biomedical survival times.

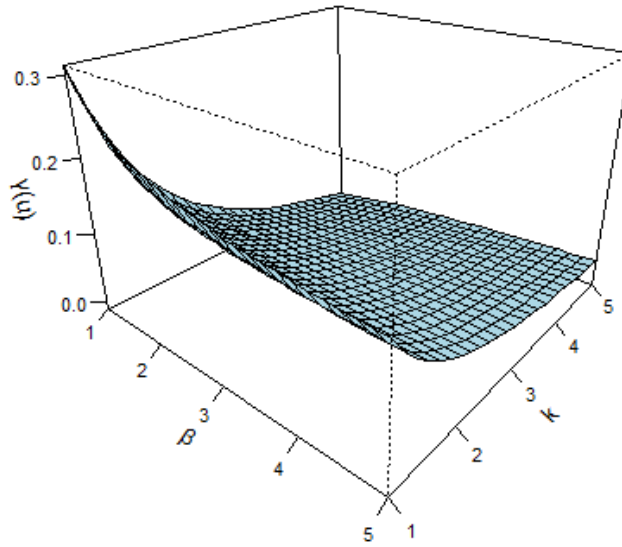


Figure 4.5 Galton's Skewness

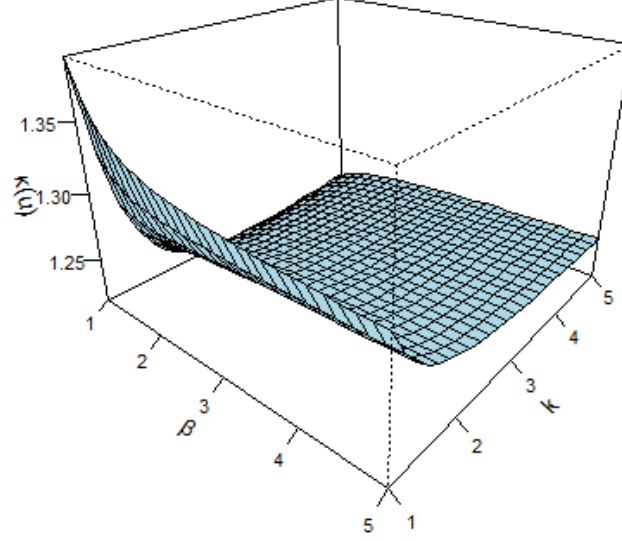


Figure 4.6 Moor's Kurtosis

4.4.3 Moments

This section derives the EPT-W distribution's non-central moment, central moment, and incomplete moment. In statistical analysis, moments are crucial for determining a distribution's central tendency, dispersion, skewness, and kurtosis.

Non-Central Moment

A non-central moment (or raw moment) of a random variable X is the expected value of its n th power, defined as;

$$\mu'_n = E[X^n] = \int_{-\infty}^{\infty} x^n f(x) dx \quad (4.19)$$

Preposition 4.3: The n th moment of the EPT-W distribution of the random variable X , is given as,

$$\mu'_n = \frac{\beta \lambda^n \Gamma\left(1 + \frac{n}{k}\right)}{1 - e^{-\alpha}} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^{i+j} \alpha^{i+1}}{i!} \binom{\beta(i+1) - 1}{j} (j+1)^{-(1+\frac{n}{k})} \quad (4.20)$$

where $\Gamma(\Phi) = \int_0^{\infty} x^{\Phi-1} e^{-x} dx$ is the Gamma function.

Proof: The series representation of the PDF expressed in equation (4.14) is

substituted in the n th non-central moment as defined in equation (4.19), we obtain,

$$\mu'_n = \int_0^\infty x^n \sum_{i=0}^\infty \sum_{j=0}^\infty \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j} x^{k-1} e^{-(j+1)(\rho x)^k} dx \quad \rho = \frac{1}{\lambda}$$

grouping like terms and simplifying,

$$\mu'_n = \sum_{i=0}^\infty \sum_{j=0}^\infty \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j} \int_0^\infty x^{k+n-1} e^{-(j+1)(\rho x)^k} dx$$

Let $a = (j+1)\rho^k$, $u = ax^k \implies x = (u/a)^{1/k}$ and $m = n+k$, also,

$$dx = \frac{1}{k} \left(\frac{1}{a} \right)^{1/k} u^{(\frac{1}{k}-1)} du$$

Which implies,

$$\mu'_n = \sum_{i=0}^\infty \sum_{j=0}^\infty \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j} \int_0^\infty \left(\frac{u}{a} \right)^{\frac{1}{k}(m-1)} e^{-u} \frac{1}{k} \left(\frac{1}{a} \right)^{1/k} u^{(\frac{1}{k}-1)} du$$

grouping like terms, factorising, and simplifying, we obtain,

$$\mu'_n = \sum_{i=0}^\infty \sum_{j=0}^\infty \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j} \frac{1}{k} \left(\frac{1}{a} \right)^{m/k} \int_0^\infty u^{(\frac{m}{k}-1)} e^{-u} du$$

by definition of the Gamma function, $\int_0^\infty u^{(\frac{m}{k}-1)} e^{-u} du = \Gamma(\frac{m}{k})$, we obtain,

$$\mu'_n = \sum_{i=0}^\infty \sum_{j=0}^\infty \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j} \frac{1}{k} \left(\frac{1}{a} \right)^{m/k} \Gamma\left(\frac{m}{k}\right)$$

substituting $m = n+k$ and $a = (j+1)\rho^k$, we get,

$$\mu'_n = \sum_{i=0}^\infty \sum_{j=0}^\infty \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j} \frac{1}{k} \left(\frac{1}{(j+1)\rho^k} \right)^{1+\frac{n}{k}} \Gamma\left(1 + \frac{n}{k}\right)$$

simplifying, we obtain,

$$\mu'_n = \frac{\beta \lambda^n \Gamma\left(1 + \frac{n}{k}\right)}{1 - e^{-\alpha}} \sum_{i=0}^\infty \sum_{j=0}^\infty \frac{(-1)^{i+j} \alpha^{i+1}}{i!} \binom{\beta(i+1) - 1}{j} (j+1)^{-(1+\frac{n}{k})}$$

where $\Gamma(\Phi) = \int_0^\infty x^{\Phi-1} e^{-x} dx$ is the Gamma function.

This completes the proof.

4.4.4 Moment-Generating Function

The moment-generating function (MGF) of a distribution, if it exists, is a special function used to compute a distribution's moments. This section shows the MGF of the EPT-W distribution.

Proposition 4.4 The MGF of the EPT-W distribution is given as,

$$M_x(t) = \frac{\beta}{1 - e^{-\alpha}} \sum_{n=0}^{\infty} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^{i+j} \alpha^{i+1} (t\lambda)^n \Gamma\left(1 + \frac{n}{k}\right)}{n! i!} \binom{\beta(i+1) - 1}{j} (j+1)^{-(1+\frac{n}{k})} \quad (4.21)$$

where $\Gamma(\Phi) = \int_0^\infty x^{\Phi-1} e^{-x} dx$ is the Gamma function.

Proof. The MGF is defined as,

$$M_x(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

using Taylor's expansion, the MGF can be rewritten as,

$$M_x(t) = \int_0^{\infty} \sum_{n=0}^{\infty} \frac{(t)^n}{n!} (x)^n f(x) dx$$

$$M_x(t) = \sum_{n=0}^{\infty} \frac{(t)^n}{n!} \mu'_n$$

substituting the nth moment of the EPT-W distribution and simplifying,

$$M_x(t) = \frac{\beta}{1 - e^{-\alpha}} \sum_{n=0}^{\infty} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^{i+j} \alpha^{i+1} (t\lambda)^n \Gamma\left(1 + \frac{n}{k}\right)}{n! i!} \binom{\beta(i+1) - 1}{j} (j+1)^{-(1+\frac{n}{k})}$$

where $\Gamma(\Phi) = \int_0^\infty x^{\Phi-1} e^{-x} dx$ is the Gamma function.

This completes the proof.

4.4.5 Rényi Entropy

Entropies are statistical measures of a random variable's uncertainty or variation. Higher entropy values indicate greater data uncertainty, and lower entropy values indicate less uncertainty.

Proposition 4.5 Let $X \sim EPTW(\alpha, \beta, \lambda, k)$ with the probability density function, (4.14), for $\delta > 0$, $\delta \neq 1$ the Rényi entropy is defined as,

$$H_\delta(X) = \frac{1}{1-\delta} \ln \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} w_{ij} \frac{1}{k} \left(\frac{1}{-(j+1)\rho^k \delta} \right)^\delta \Gamma \left(\frac{\delta(k-1)}{k} \right) \right], \quad \rho = \frac{1}{\lambda} \quad (4.22)$$

where

$$w_{ij} = \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j}$$

Proof: Rényi (1960) proposed the Rényi entropy, which is denoted as $H_\delta(X)$ for X , is defined as,

$$H_\delta(X) = \frac{1}{1-\delta} \ln \int_0^\infty f(x) dx$$

Substituting, equation (4.14) and factoring, we obtain,

$$H_\delta(X) = \frac{1}{1-\delta} \ln \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} w_{ij} \int_0^\infty x^{k-1} e^{-(j+1)(\rho x)^k} dx \right], \quad \rho = \frac{1}{\lambda}$$

Let $a = -(j+1)\rho^k$, $t = a\delta x^k$ which implies that $x = (\frac{t}{a\delta})^{1/k}$ and $dx = \frac{1}{k} (\frac{1}{a\delta})^{1/k} (t)^{\frac{1}{k}-1} dt$.

We substitute it into the $H_\delta(X)$,

$$H_\delta(X) = \frac{1}{1-\delta} \ln \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} w_{ij} \int_0^\infty \left(\left(\frac{t}{a\delta} \right)^{1/k} \right)^{k-1} e^{-t} \left(\frac{1}{k} \left(\frac{1}{a\delta} \right)^{1/k} (t)^{\frac{1}{k}-1} dt \right) \right]$$

Simplifying,

$$H_\delta(X) = \frac{1}{1-\delta} \ln \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} w_{ij} \frac{1}{k} \left(\frac{1}{a\delta} \right)^\delta \int_0^\infty t^{\frac{\delta(k-1)}{k}} e^{-t} dt \right]$$

using the Gamma function, thus

$$\int_0^\infty t^{\frac{\delta(k-1)}{k}} e^{-t} dt = \Gamma \left(\frac{\delta(k-1)}{k} \right)$$

substituting $a = -(j+1)\rho^k$, We obtain,

$$H_\delta(X) = \frac{1}{1-\delta} \ln \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} w_{ij} \frac{1}{k} \left(\frac{1}{-(j+1)\rho^k \delta} \right)^\delta \Gamma \left(\frac{\delta(k-1)}{k} \right) \right], \quad \rho = \frac{1}{\lambda}$$

where

$$w_{ij} = \frac{\beta \rho^k k (-1)^{i+j} (\alpha)^{i+1}}{i! (1 - e^{-\alpha})} \binom{\beta(i+1) - 1}{j}$$

This completes the proof.

4.4.6 Order Statistics

Statistical theory is where order statistics are most frequently encountered. Order statistics studies the properties and applications of these ordered random variables as well as functions that use them.

Proposition 4.6 Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from the EPT-W distribution with CDF, $F(x)$ and PDF, $f(x)$. We denote the order index by $r \in \{1, 2, \dots, n\}$. The CDF and PDF of $X_{r:n}$ are

$$F_{r:n}(x) = \sum_{p=r}^n \binom{n}{p} \times [F(x)]^p \times [1 - F(x)]^{(n-p)}$$

$$f_{r:n}(x) = \frac{n!}{(r-1)!(n-r)!} [F(x)]^{r-1} [1 - F(x)]^{n-r} f(x), \quad x > 0$$

Using the definitions in equations (4.10) and (4.12), we obtain,

$$\begin{aligned} F_{r:n}(x) &= \sum_{p=r}^n \binom{n}{p} \left[\frac{1 - e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right]^p \left[1 - \frac{1 - e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right]^{(n-p)} \\ f_{r:n}(x) &= \frac{n!}{(r-1)!(n-r)!} \times \left[\frac{1 - e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right]^{r-1} \times \left[1 - \frac{1 - e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right]^{n-r} \\ &\quad \times \left[\frac{\alpha \beta k \rho^k x^{k-1} e^{-(\rho x)^k} (1 - e^{-(\rho x)^k})^{\beta-1} e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right] \end{aligned}$$

Moreover, the CDF and PDF of the minimum $X_{1:n}$ are given by,

$$F_{1:n}(x) = 1 - [1 - F(x)]^n = 1 - \left[\frac{e^{-\alpha(1-e^{-(\rho x)^k})^\beta} - e^{-\alpha}}{1 - e^{-\alpha}} \right]^n$$

$$\begin{aligned} f_{1:n}(x) &= [1 - F(x)]^{n-1} f(x) = n \left[\frac{e^{-\alpha(1-e^{-(\rho x)^k})^\beta} - e^{-\alpha}}{1 - e^{-\alpha}} \right]^{n-1} \\ &\quad \times \left[\frac{\alpha \beta k \rho^k x^{k-1} e^{-(\rho x)^k} (1 - e^{-(\rho x)^k})^{\beta-1} e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right] \end{aligned}$$

and finally, the CDF and PDF of the maximum $X_{n:n}$ are given by

$$F_{n:n}(x) = [F(x)]^n = \left[\frac{1 - e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right]^n$$

$$\begin{aligned} f_{n:n}(x) &= n[F(x)]^{n-1} f(x) = n \left[\frac{1 - e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right]^{n-1} \\ &\quad \times \left[\frac{\alpha \beta k \rho^k x^{k-1} e^{-(\rho x)^k} (1 - e^{-(\rho x)^k})^{\beta-1} e^{-\alpha(1-e^{-(\rho x)^k})^\beta}}{1 - e^{-\alpha}} \right] \end{aligned}$$

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